# Pool Ball Detection and Recognition

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**Abstract**

This report documents the experimental development of a two-stage computer vision pipeline for localising and recognising pool balls. Stage 1 compared a green-channel boost, fixed HSV and single-channel lightness thresholding, and finally adopted multi-channel CIELAB edge fusion with a dynamic pocket mask, physical NMS, geometric cushion constraints, and a raw-frame Hough recovery path projected through the table homography. Stage 2 compared K-Means colour clustering and finally adopted one-dimensional hue and saturation histograms with a reduced circular sampling mask, a 30/70 saturation-dominant weighting, and a split-mask cue gate. Across the six test tables the final pipeline achieves Stage 1 precision 0.911, recall 0.911 and F1-score 0.911, with Stage 2 recognition accuracy 76.8% on correctly localised balls.

## 1 Introduction

The task was to localise every pool ball on a rectified table image and then assign each surviving detection to one of the sixteen known balls. The six test images contain strong shadows, uneven lighting, tightly packed racked balls, and a mixture of solids and stripes. I developed the final system through a sequence of explicit experiments, and I retained a detailed error analysis because the remaining failures reveal the limits of the chosen detectors and descriptors. The project repository is available at https://github.com/Eskndrani/OpenCV-Pool-Ball-Tracker.

### 1.1 End-to-End Pipeline Overview

Each image is rectified into the $1434 \times 683$ metric frame. Stage 1 fuses independent Canny edges from the $L/a/b$ channels, applies a dynamic pocket and netting mask, and runs Hough Circles with a 40-pixel minimum separation plus cushion (22 px) and NMS (40 px) filtering. In parallel, a secondary recovery path runs standard integrated Canny edge detection directly on the unwarped camera frame before raw-frame Hough Circles; its candidates are projected into metric space through the homography $\mathbf{H}$. Stage 2 evaluates survivors using 30/32-bin hue and saturation histograms, the split-mask cue gate, the 30/70 Bhattacharyya distance, and a 0.55 rejection threshold (Figure 1).

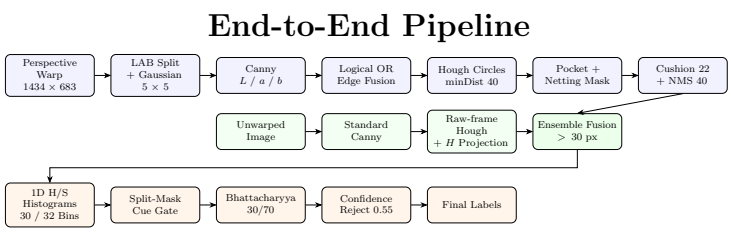



Figure 1: End-to-end pipeline architecture.

# 2 Stage 1: Ball Detection

## 2.1 Perspective rectification

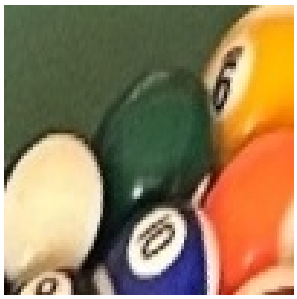
Each input photograph is first mapped by a planar homography from the four measured cushion-to-cushion corners into a $1434 \times 683$ metric table frame. The destination dimensions match the physical $1434\,\text{mm} \times 683\,\text{mm}$ playing area, so one transformed pixel corresponds to one millimetre. This metric frame is used by all later stages.

## 2.2 Failed green-channel boost and adaptive thresholding

To solve the problem of missing green balls, I initially hypothesised that isolating and boosting the RGB green channel by a factor of 1.5, followed by adaptive Gaussian thresholding, would force the balls to pop against the felt. The adaptive threshold used a local block size of 15 and a constant of 2. This failed because boosting the green channel equally boosted the green table, and adaptive thresholding amplified the woven texture of the felt into severe noise. The original region, green-boosted grayscale image, and adaptive threshold map were inspected together. The ball boundaries are washed out while the felt texture dominates the edge map.

## 2.3 Failed HSV and single-channel lightness thresholding

I then attempted to isolate the felt directly in HSV space with a fixed hue range of 35–85, saturation range 40–255, and value range 40–255. This also failed for the green object balls, particularly balls 6 and 14, because their hue and saturation overlap almost completely with the green felt. A $100 \times 100$ region around the green ball and the resulting binary mask were inspected. In the mask, the green ball vanishes into the white felt instead of producing a distinct foreground blob. A single-channel lightness threshold produced the complementary failure: shadows fragmented the ball boundaries and the threshold could not separate dark balls from the table. Both results motivated a multi-channel approach.



(a) Region around the green ball.



(b) Fixed HSV mask: green ball merges into felt.

Figure 2: Static HSV thresholding fails for green object balls: the green ball vanishes into the green felt instead of forming a distinct foreground blob.

## 2.4 Successful multi-channel CIELAB edge fusion

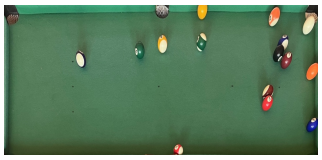
I realised that the solution required looking at colour boundaries, not just brightness. I split the rectified image into the CIELAB colour space and applied Gaussian smoothing with a $5 \times 5$ kernel and a standard deviation of 1.2 to all three channels. I then ran Canny edge detection independently on each channel. I used thresholds of 45 and 95 for the

lightness channel $L$, and more sensitive thresholds of 35 and 85 for the chromatic channels $a$ and $b$. The three binary edge maps were combined with a logical OR to produce a highly robust gradient map that captured subtle colour shifts as well as luminance boundaries.

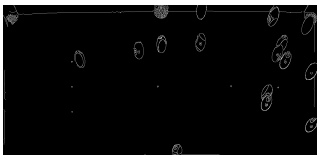
**Key Idea: LAB Multi-Channel Edge Fusion**

The lightness channel supplies the strong outer silhouette of each ball, while the chromatic channels supply the colour boundaries that remain visible when a ball is dark, shadowed, or similar in brightness to the felt. Fusing the three edge maps produces a gradient image that is markedly more complete than any single-channel edge map.

Figure 3 shows the rectified Table 4 image and the corresponding multi-channel LAB edge map used by the circular detector.



(a) Rectified Table 4.



(b) Gaussian-smoothed LAB Canny edge map.

Figure 3: Rectification and multi-channel LAB Canny edge fusion.

## 2.5 Dynamic pocket mask, netting mask, and physical NMS

The fused edge map is passed to the Hough Circle Transform with a minimum centre separation of 40 pixels and a ball-radius range of 19 to 24 pixels. I initially rejected every candidate within a blind 95-pixel radius of the six theoretical pockets, but visual debugging showed that this removed genuine balls near the corners, including the bright cue ball beside the top-right pocket of Table 4. I therefore replaced the blind radius with a dynamic pocket mask.

For each of the six pocket centres, I examined a $120 \times 120$ pixel region on the raw lightness channel and thresholded two complementary structures: the dark pocket holes at $L < 35$ and the bright metallic brackets at $L > 200$. The mid-gray pocket netting is not captured by either threshold, so I also extracted the combined LAB edge map in the same region and dilated it with a $15 \times 15$ circular kernel. The netting is a dense web of edges, so this dilation fuses it into a solid morphological block. The fused netting block was OR-ed with the L-extreme mask. The local mask was restricted to a 70-pixel radius around the pocket centre so that a bright ball near a pocket would not be mistaken for a bracket, and the six local masks were combined and dilated with a $13 \times 13$ circular kernel to cover the immediate pocket lip. A Hough candidate is rejected only when its centre lies on this complete hole–bracket–netting mask. The cushion exclusion margin remains at 22 pixels. An experiment reducing it to 5 pixels recovered far-edge balls but reintroduced rail shadows that bypassed the Stage 2 filter, so the larger margin was restored.

I observed that Hough occasionally returned intersecting accumulator peaks that described the same physical ball. My first response was a strict 40-pixel physical NMS, which retains the strongest peak in accumulator order and suppresses any later centre within the physical impossibility radius. I also measured the fused edge map: it contained 1.025 pixels per skeleton pixel, so Zhang–Suen thinning removed almost nothing and did not eliminate the ghost responses. I then swept the accumulator threshold and found that ghosts disappeared only at a threshold of 14, while the number of true balls fell from 83 to

15. No clean accumulator threshold exists for this dataset. I therefore implemented a secondary raw-frame path: I run LAB Canny and Hough on the unwarped photograph ($param2 = 14$, minimum separation 30, radius 18–30), project each centre through the homography into metric coordinates, and append only candidates farther than 30 pixels from every rectified detection. This recovered three true balls, raising TP from 79 to 82 and reducing FN from 11 to 8 with FP unchanged, so F1 rose from 0.893 to 0.911 and Stage 2 accuracy from 75.9% to 76.8%.
The dynamic mask covers the dark hole, bright bracket, and netting, while the cue ball is spared and recovered by the final detector.

# 3 Stage 2: Ball Recognition and Classification

## 3.1 Failed K-Means clustering

To differentiate striped from solid balls, I first attempted to use K-Means clustering with $K = 2$ on the pixels inside each ball mask to extract two dominant colours, then compared those colours with the reference balls using Euclidean distance. This proved computationally heavy and unreliable because the visibility of the white stripe depends on the physical rotation of the ball. On a solid blue ball, the two clusters split the number and the specular reflections rather than producing a stable stripe-versus-solid descriptor.

## 3.2 Successful saturation-dominant histograms

I pivoted to one-dimensional histograms. I extracted a hue histogram with 30 bins and a saturation histogram with 32 bins from a reduced circular sampling mask of radius 16 pixels. The reduced mask prevents a slightly off-centre detection from admitting felt, shadow, or corner pixels into the descriptor. I then compared test histograms with the sixteen reference histograms using the Bhattacharyya distance. Because a white stripe drastically lowers a ball's average saturation regardless of rotation, I applied a custom weighting system: 30% weight to hue, 70% weight to saturation, and 0% weight to brightness.

**Saturation-Dominant Distance**

$$d = 0.30\, d_{\text{hue}} + 0.70\, d_{\text{sat}}. \tag{1}$$

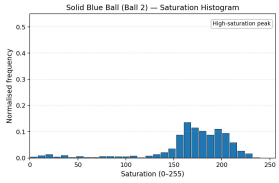
A detection whose weighted distance exceeds 0.55 is rejected as a non-ball or as an unreliable observation.

I observed that Stage 1 occasionally returned circular shadow artifacts along the rails. Tightening the Bhattacharyya rejection threshold to 0.55 removed them, because shadows lack the sharp saturation peaks of resin balls. I found that the cue histogram overlapped false positives because of green-felt reflections, so I implemented a split-mask gate: a 16-pixel centre for brightness and whiteness, and a full 21-pixel mask for overall paint. My initial thresholds admitted the purple 12 stripe; after I tightened the centre value above 235, centre saturation below 75, and full saturation below 76, the purple stripe and the 9-ball were excluded while the cue was retained. I use the Highlander rule to select the lowest full saturation as the cue; all other candidates fall back to their best non-cue histogram match.
In Figure 4 I compare the 1D saturation distributions of solid blue ball 2 and striped blue ball 10. The solid ball peaks at high saturation, whereas the striped ball has a pronounced low-saturation peak caused by its white stripe. I found this shift independent of the ball's

physical rotation, which is why the saturation channel provides a mathematically robust feature for distinguishing stripes from solids.



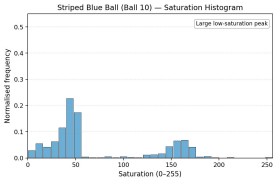
(a) Solid ball 2 crop.




(b) Solid ball's high-saturation peak.



(c) Striped ball 10 crop.




(d) Striped ball's low-saturation peak.

Figure 4: Blue-ball comparison: the solid ball 2 crop and its high-saturation histogram (left pair), and the striped ball 10 crop with its low-saturation peak caused by the white stripe (right pair).

# 4 Results and Error Analysis

## 4.1 Stage 1: Detection Results

Stage 1 metrics use strict one-to-one greedy matching between the final drawn detections and the ground-truth centres. A detected centre $(x_{\text{det}}, y_{\text{det}})$ is a True Positive if its Euclidean distance to an unmatched ground-truth centre $(x_{\text{gt}}, y_{\text{gt}})$ satisfies

$$\sqrt{(x_{\text{det}} - x_{\text{gt}})^2 + (y_{\text{det}} - y_{\text{gt}})^2} \leq 21.5 \text{ px},$$

where 21.5 px is half the 43 px ball diameter. Any detection left unmatched is a False Positive, and any ground-truth centre left unmatched is a False Negative. Precision, recall and F1 are

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad \text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \quad \text{F1} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}.$$

Table 1 reports the resulting metrics: overall precision is 0.911, recall is 0.911, and the F1-score is 0.911.

Table 1: Stage 1 detection metrics after dynamic pocket masking.

| Image | TP | FP | FN | Precision | Recall | F1-score |
|---|---|---|---|---|---|---|
| Table 1 | 13 | 0 | 3 | 1.000 | 0.812 | 0.897 |
| Table 2 | 15 | 2 | 1 | 0.882 | 0.938 | 0.909 |
| Table 3 | 16 | 0 | 0 | 1.000 | 1.000 | 1.000 |
| Table 4 | 14 | 2 | 1 | 0.875 | 0.933 | 0.903 |
| Table 5 | 13 | 2 | 1 | 0.867 | 0.929 | 0.897 |
| Table 6 | 11 | 2 | 2 | 0.846 | 0.846 | 0.846 |
| Overall | 82 | 8 | 8 | 0.911 | 0.911 | 0.911 |

## 4.2 Stage 2: Recognition Results

Stage 2 recognition accuracy is evaluated strictly over the localized Stage 1 True Positives:

$$\text{Recognition Accuracy} = \frac{\text{correctly labelled Stage 1 TPs}}{\text{total Stage 1 TPs}}.$$

A Stage 1 TP is counted as incorrect when its 30/70 weighted hue–saturation Bhattacharyya distance is closest to a different reference ball, or when the confidence filter ($d > 0.55$) rejects it. Across all six tables, 63 of 82 correctly localised balls receive the correct label, giving an overall recognition accuracy of 76.8%. Figure 5 shows the final labelled output for Table 4, which combines high Stage 1 recall with 92.9% Stage 2 recognition accuracy.

Table 2: Stage 2 recognition accuracy on correctly localised balls.

| Image | Correct labels | True positives | Accuracy |
|---|---|---|---|
| Table 1 | 10 | 13 | 0.769 |
| Table 2 | 10 | 15 | 0.667 |
| Table 3 | 12 | 16 | 0.750 |
| Table 4 | 13 | 14 | 0.929 |
| Table 5 | 10 | 13 | 0.769 |
| Table 6 | 8 | 11 | 0.727 |
| Overall | 63 | 82 | 0.768 |

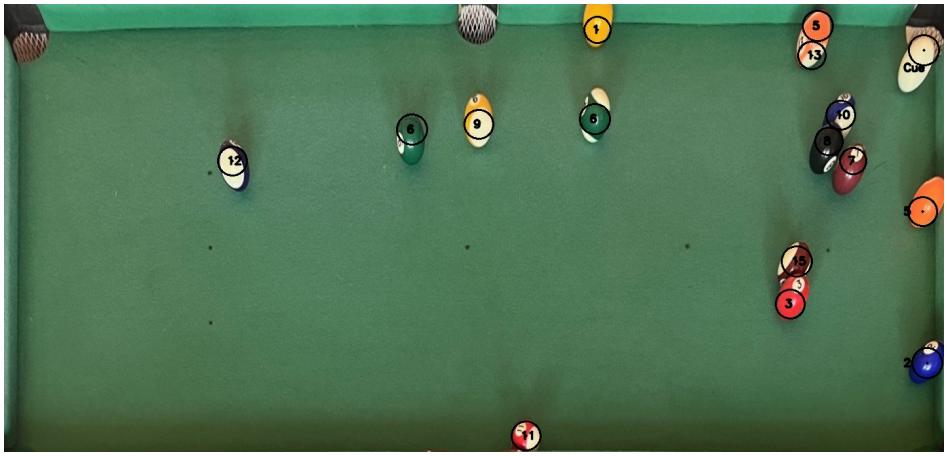


Figure 5: Final labelled output for Table 4 after dynamic pocket masking and Stage 2 classification; Stage 2 recognition accuracy is 92.9%.

## 4.3 Error Analysis

After restoring the 22-pixel cushion margin, applying the split-mask cue gate, and fusing rectified with raw-frame candidates, Stage 1 yields 82 true positives, 8 false positives, and 8 false negatives; Stage 2 reaches 76.8% recognition accuracy. I audited the Table 2 overlay against the final ensemble metrics because it isolates the failures that remain active: duplicate detections on a single perimeter ball, missed interior balls in the right-hand cluster, and rotation/specular misclassification. The cue and the 9-ball are now resolved, so I no longer cite a cue-gate failure. Figure 6 shows the overlay.

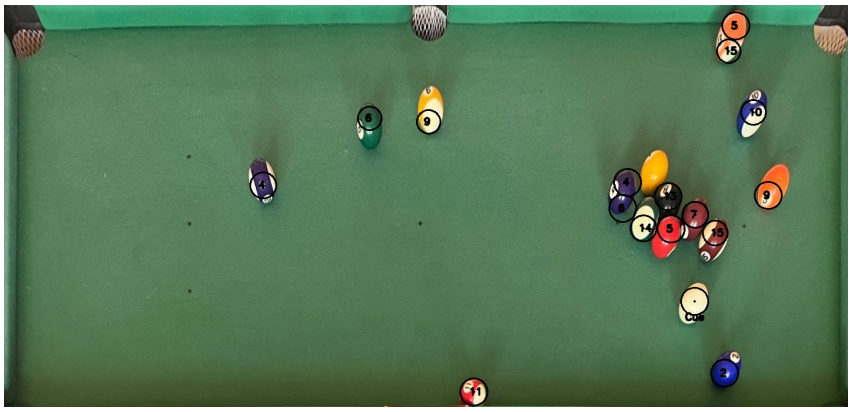


Figure 6: Final labelled output for Table 2, illustrating duplicate detections on single perimeter balls (top right) and missed detections inside dense clusters; no elliptical shifting occurs.

**Failure Mode 1: Duplicate Detections on Single Balls.** In my evaluation I observed that candidates near a table boundary can cast two accumulator peaks for the same physical ball. The cushion truncates the perimeter arc, and the two surviving arcs on opposite sides of the ball can produce centres more than 40 pixels apart, so they lie just outside the 40-pixel physical NMS and both survive. Figure 6 shows this at the top right of Table 2, where a single ball receives two overlapping detection circles labelled 5 and 15. One is the correct localization and the other is an artificial false positive; there is no elliptical centre distortion involved, only a duplicated circular response.

**Failure Mode 2: Dense-Cluster Gradient Nullification.** In the right-hand cluster of Table 2 I observed that touching balls break the continuous Canny edge arcs that the circular Hough transform requires. The outer boundaries of adjacent balls merge into one irregular gradient, so the accumulator receives spread votes and the peak for an interior ball stays below the $param2 = 8.0$ threshold. Raising the threshold or enlarging the radius range does not recover those balls without admitting shadow arcs, so I kept the stricter setting. This is the source of the remaining false negative in Table 2: the interior ball is visible to me but has no isolated circular evidence for Hough to fit. A Watershed segmentation pass before circle fitting would be the natural remedy.

**Failure Mode 3: Rotation and Specular Misclassification.** My evaluation showed that rotation and specular reflections can shift a ball's 1D hue/saturation histogram bins even when its centre is localized correctly. A white number circle or a bright highlight can dominate the saturation peak, while a shadow can suppress the paint peak; the 30/70 weighted Bhattacharyya distance then lands closest to the wrong reference or exceeds the 0.55 rejection threshold. Table 2 exposes this in its Stage 2 accuracy of 66.7%, where correctly localized balls such as the 3, 8, 12, and 13 are assigned neighbouring labels or pulled toward the cue reference. I kept the deterministic histogram classifier because the errors are sparse, but a specular-rejection mask or a rotation-invariant stripe descriptor would be the next refinement.

# 5 Discussion

The final Stage 1 precision is 0.911, recall is 0.911, and F1 is 0.911; Stage 2 accuracy is 0.768. I attribute this balance to two design choices. First, multi-channel LAB edge fusion gave the detector complementary evidence: the lightness channel supplied the

strong outer silhouette, while the $a$ and $b$ channels recovered colour boundaries that would otherwise vanish in shadow. The cost of this fusion was a denser edge map, which produced offset accumulator arcs; the 40-pixel physical NMS and the 22-pixel cushion margin were therefore not optional refinements but necessary consequences of the richer edge evidence. I also found that the rectified-only detector consistently under-counted perimeter balls, which is why I added the raw-frame recovery path. That path recovered three true balls without adding false positives, but it also showed that a lower Hough threshold alone could not improve recall without a large precision penalty.

For recognition, I found that 1D saturation histograms were substantially more robust than K-Means colour clustering. The 30/70 hue–saturation weighting encodes the physical fact that a white stripe lowers saturation independently of rotation, while the split-mask cue gate prevents a bullseye 9 or 14 from hijacking the cue label when its white number circle faces the camera. The remaining limits are duplicate perimeter responses, dense-cluster merging, and specular/stripe overlap under rotation. I believe a spatial stripe descriptor and a Watershed segmentation pass are the most promising next steps, because both would add shape evidence that the current histogram-only Stage 2 cannot access.

# 6 Conclusion

In this project I implemented a two-stage pipeline that rectifies the table, detects candidate balls with multi-channel LAB edge fusion and Hough circles, recovers perimeter balls with a raw-frame secondary path, and classifies survivors with saturation-dominant histograms and a split-mask cue gate. The final ensemble achieves Stage 1 precision 0.911, recall 0.911, and F1 0.911, with Stage 2 recognition accuracy 0.768.

My experiments showed that LAB edge fusion was essential for shadowed and green balls, but the resulting edge density made the 40-pixel NMS and 22-pixel cushion margin necessary to suppress offset arcs. The raw-frame recovery pass added three true balls without increasing false positives. For recognition, the 30/70 hue–saturation histogram was more robust than K-Means, and the split-mask gate prevented bullseye stripes from hijacking the cue label. Residual errors are duplicate perimeter responses, dense-cluster edge merging, and specular/stripe histogram confusion; a spatial stripe descriptor and Watershed segmentation are the natural next steps.

# AI Usage Statement

Generative AI tools (Google Gemini) were used during this project to assist with report text revisions, LaTeX/TikZ diagram formatting, and generating standalone C++ diagnostic scripts for parameter sweeps.

**Prompt & Response Example 1 (Parameter Sweep Script):** *User Prompt:* "Write a diagnostic script to sweep Hough param2 from 6 to 18 and compare edge-thinned LAB maps with raw Canny maps." *AI Response:* "Generated a C++ benchmark logging TP/FP counts to stage2_feature_analysis.csv."

**Prompt & Response Example 2 (Gate Suggestion):** *User Prompt:* "The 1D histogram labels the 9-ball as the cue when its white circle faces the camera. How can I separate them?" *AI Response:* "Suggested a split-mask saturation/value check comparing the 16px centre with the 21px full ball."

**Prompt & Response Example 3 (LaTeX Troubleshooting):** *User Prompt:* "The pipeline TikZ block wraps awkwardly and overflows the margin. How do I format a two-stage flow?" *AI Response:* "Provided matrix/node code with explicit node distances and rounded styles."

All core algorithmic implementations, experimental designs, parameter selections, and final data verification were conducted directly by the author.